Practice workbook · Name joint types and compare their movement

What a joint is, hinge, ball-and-socket and gliding joints, where they occur and the movement each allows.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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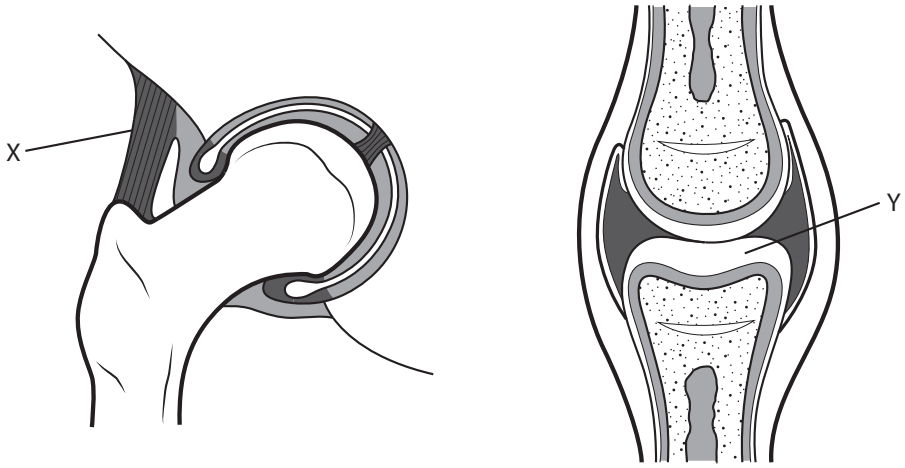
Type ID: HBI-JOINT-TYPE-MOVEMENT

1. Question 5 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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1. Question 5 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

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(ii) Give the role of part Y.

(1)

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(Total for Question 5 = 6 marks)

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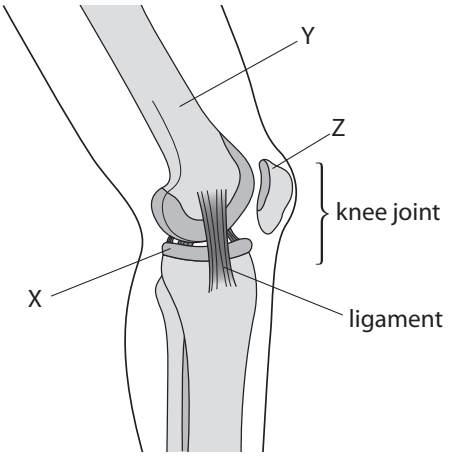
Working and answer

2. Question 1 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

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2. Question 1 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

(b) Aerobic respiration releases energy for muscles to contract.

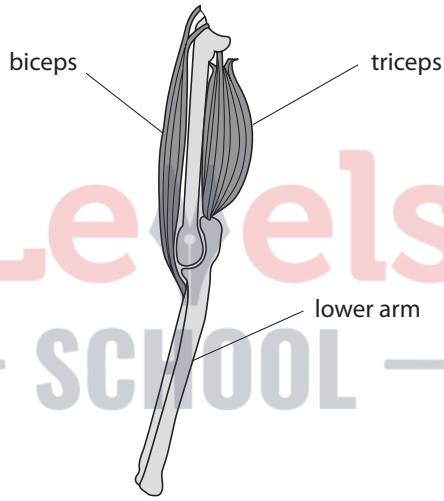
Complete the word equation for aerobic respiration.

(2)

glucose + ... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

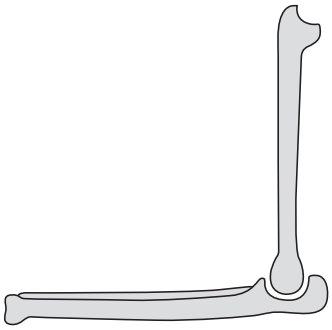
Diagram 1



Complete diagram 2 to show how the muscles change when the lower arm is raised.

(2)

Diagram 2



(Total for Question 1 = 9 marks)

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2. Question 1 · part a.iii · 1 mark · Working and answer

Working and answer

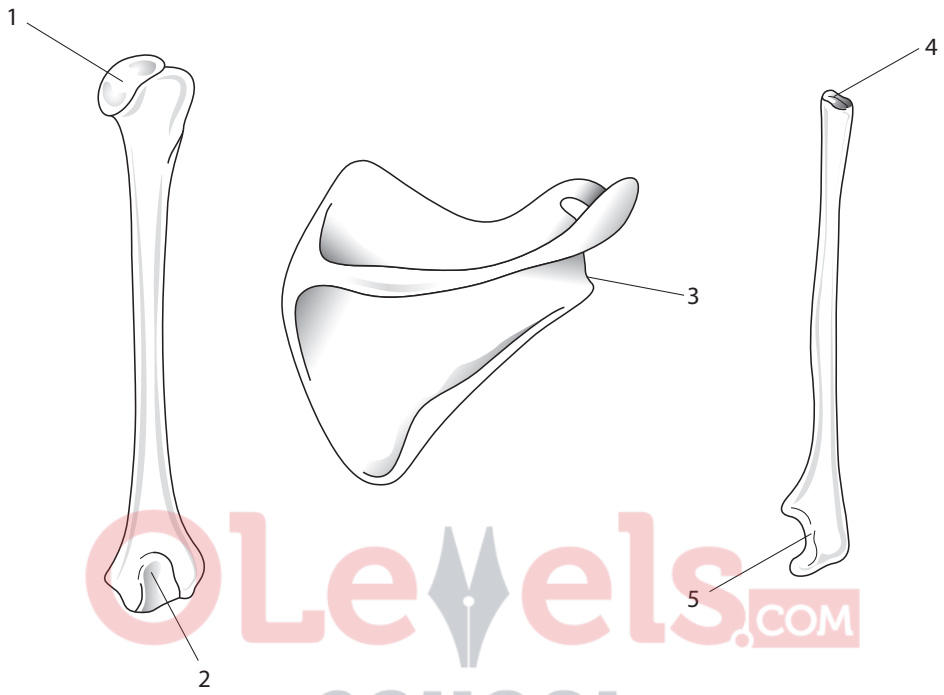
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3. Question 2 · part a · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

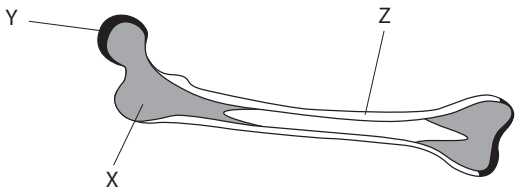
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3. Question 2 · part a · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

.....

(iv) Describe the function of the part labelled Y.

(2)

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3. Question 2 · part a · 1 mark · Working and answer

Working and answer

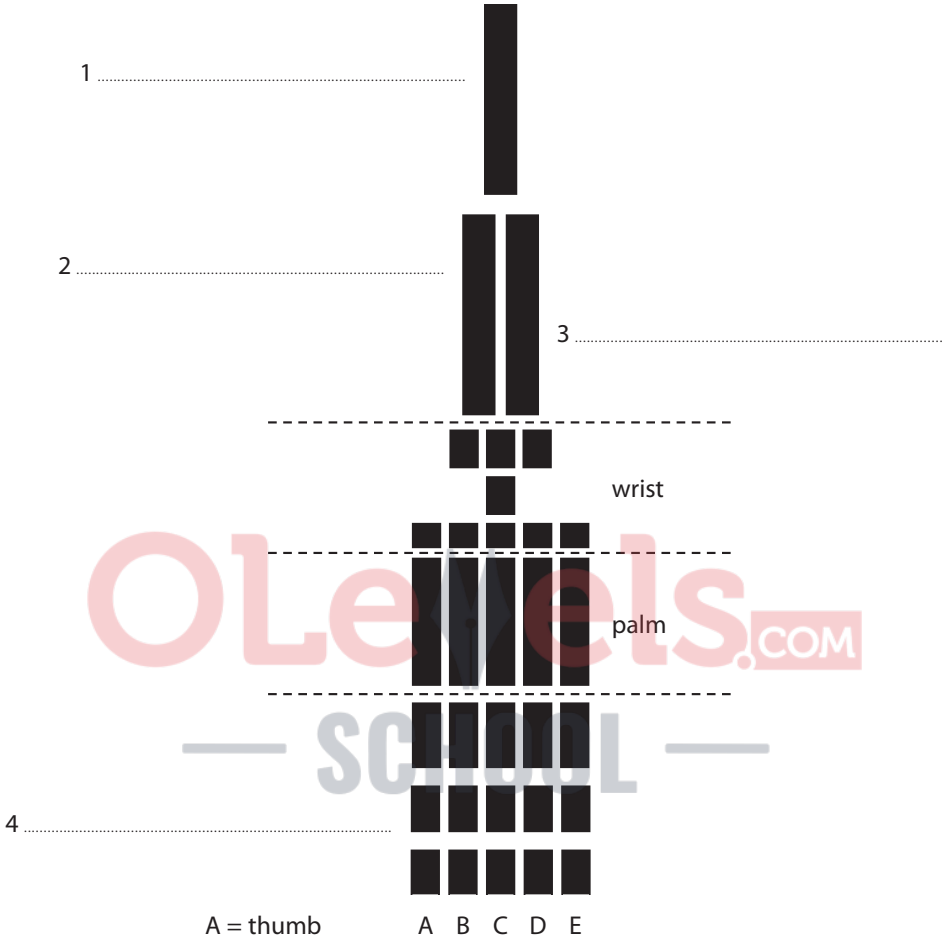
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4. Question 2 · part a.ii · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.ii only.

2 The diagram shows a model representing the arrangement of bones in the arm.



(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

(4)

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

(1)

(iii) Name two other types of synovial joint found in the skeleton.

(2)

1 .....

2 .....

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4. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

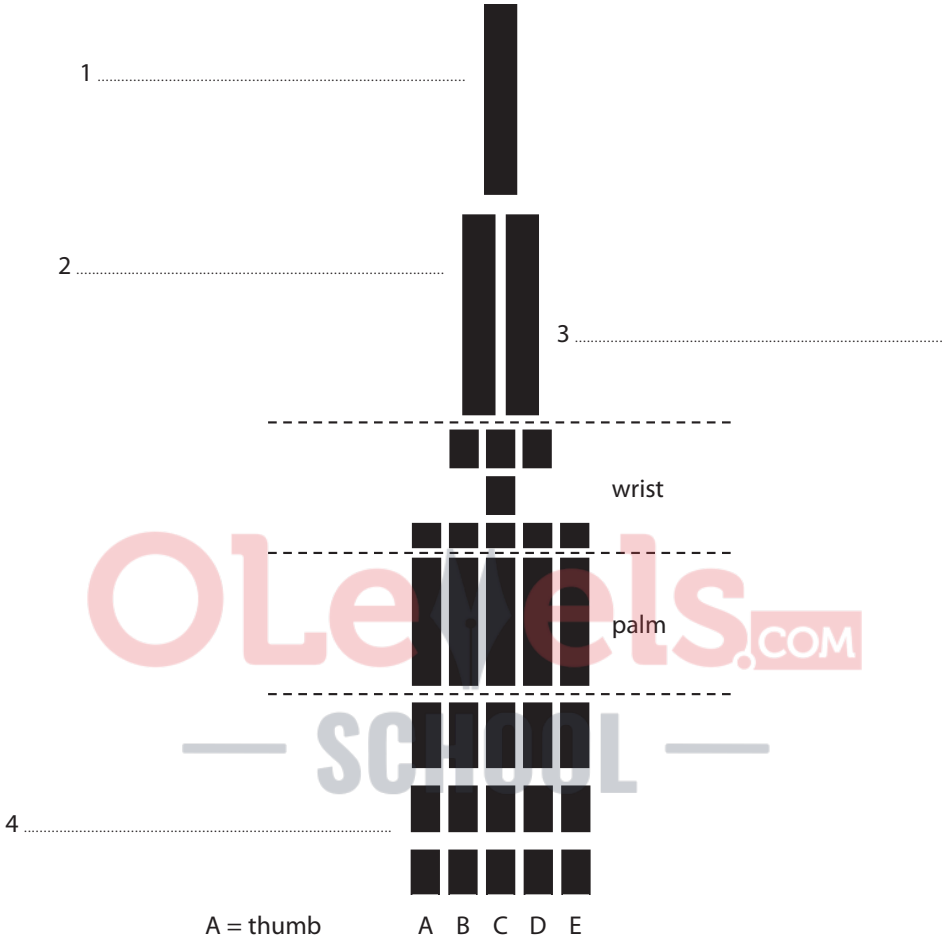
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5. Question 2 · part a.iii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

2 The diagram shows a model representing the arrangement of bones in the arm.



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(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

(4)

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

(1)

(iii) Name two other types of synovial joint found in the skeleton.

(2)

1 .....

2 .....

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5. Question 2 · part a.iii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

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(b) Vitamin D is important for the growth and development of healthy bones.

Explain how vitamin D helps the growth and development of bones.

(3)

(Total for Question 2 = 10 marks)

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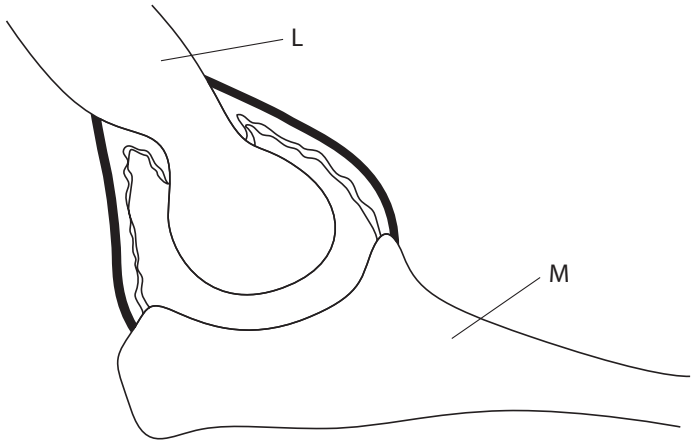
Working and answer

6. Question 6 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

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(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

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6. Question 6 · part a · 2 marks · Working and answer

Working and answer

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7. Question 2 · part a · 2 marks

June 2023 Paper 1R

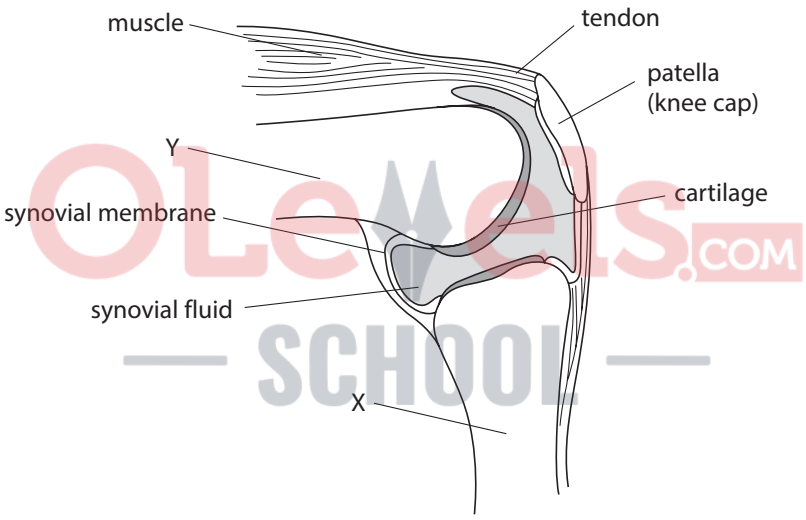
Further practice · Attempt part a only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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7. Question 2 · part a · 2 marks · Working and answer

Working and answer

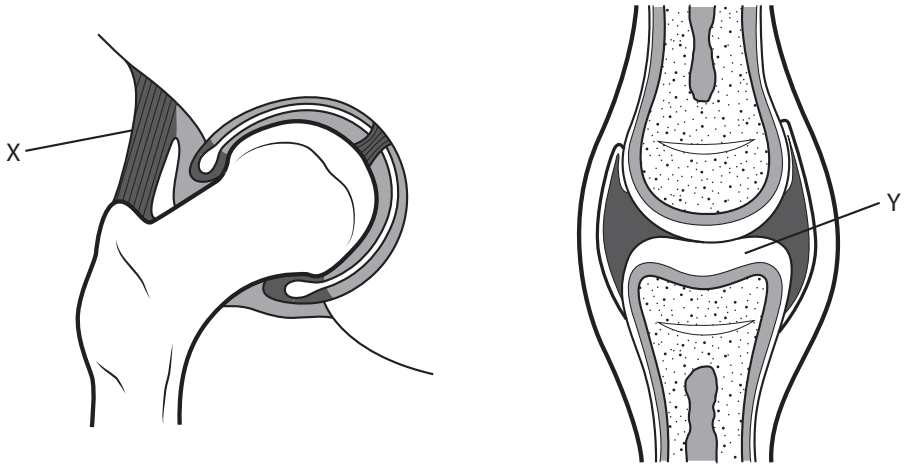
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8. Question 5 · part a.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.i only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B.

(2)

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B.

(2)

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8. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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